

Our Environment

Eco-system — what are its components?

- All the interacting organisms in an area together with the non-living constituents of the environment form an ecosystem.
- an ecosystem consists of biotic components comprising living organisms and abiotic components comprising physical factors like temperature, rainfall, wind, soil and minerals.

Food Chains and Webs

- series of organisms taking part at various biotic levels form a food chain
- Each step or level of the food chain forms a trophic level.
- The autotrophs or the producers are at the first trophic level. They fix up the solar energy and make it available for heterotrophs or the consumers.
- The herbivores or the primary consumers come at the second, small carnivores or the secondary consumers at the third and larger carnivores or the tertiary consumers form the fourth trophic level

Ozone Layer and How it is Getting Depleted

- Ozone (O₃) is a molecule formed by three atoms of oxygen.
- While O₂, which we normally refer to as oxygen, is essential for all aerobic forms of life.
- Ozone, is a deadly poison.
- at the higher levels of the atmosphere, ozone performs an essential function.
- It shields the surface of the earth from ultraviolet (UV) radiation from the Sun.
- This radiation is highly damaging to organisms, for example, it is known to cause skin cancer in human beings.
- Ozone at the higher levels of the atmosphere is a product of UV radiation acting on oxygen (O₂) molecule.
- The higher energy UV radiations split apart some molecular oxygen (O₂) into free oxygen (O) atoms.
- These atoms then combine with the molecular oxygen to form ozone as shown—
The amount of ozone in the atmosphere began to drop sharply in the 1980s.
- This decrease has been linked to synthetic chemicals like chlorofluorocarbons (CFCs) which are used as refrigerants and in fire extinguishers.
- In 1987, the United Nations Environment Programme (UNEP) succeeded in forging an agreement to freeze CFC production at 1986 levels.